Rule based English-Vietnamese bilingual terminology extraction from Vietnamese documents

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ABSTRACT

Bilingual terminologies are important resources for natural language processing as well as for human use. The automatic acquisition of bilingual terminologies is mostly based on bilingual corpora. However, monolingual corpora could also be a good source for extracting bilingual terms. In fact, as English is used for international publications, we often find English words together with their translation in monolingual documents written in other languages. In this paper, we propose a rule-based approach for automatically extracting English-Vietnamese bilingual terminology from Vietnamese monolingual documents. Experiments on a medical corpus and on Wikipedia documents show that the extracted bilingual term candidates are of high quality.

KEYWORDS

bilingual term, bilingual terminology extraction, English-Vietnamese bilingual term, mutual information, rule based, term extraction

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1 INTRODUCTION

Terms are single words or multi-words associated to a specific domain. Bilingual terminologies are important resources, as exploiting information of bilingual terms brings many benefits not only for human cross-lingual use, but also for solving problems related to natural language processing (NLP). For example, bilingual terminology helps to significantly increase the quality of machine translation, as terms will be translated more accurately and more contextually [3] [8]. Another problem is language mixing in documents, which increases the difficulty for text processing and text understanding. In social network texts, the use of English terms intermingled in Vietnamese texts is becoming more and more common, such as the equivalent use of "chụp cắt lớp" and "CT scan", "bình luận" and "comment", *etc.* Bilingual term resources bring many obvious benefits in handling this issue. They are also precious tools for cross-language information retrieval.

The automatic acquisition of bilingual terminologies is mostly based on bilingual corpora. Several methods have been proposed for extracting bilingual terms from parallel or comparable corpora, as in [9] or [11]. However, monolingual corpora could also be a good source for extracting bilingual terms. In fact, as English is used for international publications, we often find English words together with their translation in monolingual documents written in other languages. For low-resource languages like Vietnamese, mining these monolingual documents allows us to extract bilingual terms, especially new terms in specialized domains. A similar idea in this direction is found in the work of Bond et al. [4] for extracting Japanese-English and Chinese-English bilingual terminologies.

In this paper, we present a method to extract an English-Vietnamese bilingual terminology from Vietnamese monolingual documents. We focus on extracting Vietnamese terms and term explanations in English written in parentheses immediately after these terms. To do that, we scan the documents to extract candidate English terms, then propose a set of rules for extracting and filtering bilingual term candidates. Experiments are conducted on two corpora: a medical corpus and the Wikipedia corpus in Vietnamese.

The remaining of the paper is organized as follows: Section 2 introduces the related works for bilingual term acquisitions, Section 3 presents our proposed method, and Section 4 shows the obtained result, before the conclusion in Section 5.

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2 RELATED WORKS

As we mentioned above, bilingual terminologies are important for NLP as well as for human use. Therefore, many works carried on the subject of bilingual terminology extraction have been published. We introduce here various approaches adopted to extract bilingual terms in the past ten years.

Ha et al. [9] propose a methodology to perform bilingual terminology extraction by combining the monolingual terminology extraction for one language and an available mechanisms for aligning noun phrases in a parallel corpus. Experimenting for the English-Spanish language pair, the authors show that this technique helps to improve the quality of term extraction in each language.

Saralegi et al. [15] propose a method for extracting English-Basque term pairs using context similarity and cognates (words with similar forms).

Exploiting the idea of extracting bilingual terms from mainly (or partly) monolingual documents, Bond et al. [4] propose a new method to raise the precision of such an extraction system by exploiting explicit in-text cues, concretely the use of parentheses "()" and character type (for Chinese and Japanese language). This allows to greatly improve the precision of previously proposed systems for Japanese-English and Chinese-English term extraction, from under 20% to 73% of useful bilingual terms.

In [11], Els Lefever *et al* introduce a language-pair independent terminology extraction module based on a sub-sentential alignment system that links linguistically motivated phrases in parallel texts. Statistical filters are applied on the bilingual term candidates extracted from the alignment output.

Guinovart and Simões [7] present another parallel corpora-based bilingual terminology extraction method based on the occurrence of bilingual morpho-syntactic patterns in probabilistic translation dictionaries.

In [14], Sadat exploits the idea of using multilingual web-based encyclopedias such as Wikipedia as comparable corpora for bilingual terminology extraction, making use of Wikipedia link information and data. The method was experimented with Japanese, English and French, showing the possibility to work on different language pairs.

Zhang and Wu [18] present a bilingual term alignment method based on the difference in distribution of general words and terminologies.

In 2012, Aker et al. [1] publish a method for extracting parallel phrases from comparable news articles. Their approach consists of a phrase pair generator that automatically generates candidate parallel phrases and a binary SVM classifier that classifies the candidate phrase pairs as parallel or non-parallel.

Also based on parallel text alignment, in 2013 Macken et al. [13] use a sophisticated chunk-based alignment method for the generation of candidate terms, after which the specificity of the candidate terms is determined by combining several statistical filters.

Arcan et al. [2] exploit cross-lingual links in Wikipedia to extract bilingual terms for use in statistical machine translation system.

In [5], Gaizauskas *et al* describe a multi-component system called BiTES (Bilingual Term Extraction System) designed to automatically gather domain-specific bilingual term pairs from Web data. BiTES components consist of data gathering tools, domain classifiers, monolingual text extraction systems and bilingual term aligners.

Yang et al. [17] work on a Chinese-Japanese parallel corpus of patents. They automatically extract multi-word terms from monolingual corpora by combining statistical and linguistic filtering methods, then use the sampling-based alignment method to identify aligned terms to select the most promising bilingual multi-word terms.

In [10], Sanjika Hewavitharana and Stephan Vogel propose a method that consists in extracting parallel phrases from comparable data. In this method, they explore three phrase alignment process to detect parallel phrase pairs embedded in comparable sentences: a standard phrase extraction algorithm relying on the Viterbi path; a phrase extraction module using only lexical features; and a binary classifier to detect parallel phrase pairs among a large collection of phrase pair candidates.

In [6], Shengxiang Gao *et al* present a method to extract Chinese-Vietnamese bilingual metallurgy terms based on a pivot language. They first train a model to identify and extract Chinese metallurgy terminology. Then, a phrase-based statistical machine translation model is used to generate the Chinese-English phrase table and English-Vietnamese phrase table in the parallel multilingual corpus, allowing to infer a Chinese-Vietnamese phrase table. Finally, a Chinese-Vietnamese bilingual metallurgy term base is built by combining the former extracted Chinese metallurgy terms and the inferred phrase table.

In [12], Jingshu Liu *et al* propose a unified framework for aligning bilingual terms independently of the term lengths to extract bilingual terminologies from parallel corpus, by improving different state-of-the-art approaches both for single-word term and multi-word term alignements by exploiting bilingual word embedding approach.

In summary, most known approaches for bilingual terms extraction make use of parallel or comparable corpora. However, the availability of such multilingual data is not always trivial. In this paper, we explore an idea similar to that of Bond et al. [4], which is to find the equivalent Vietnamese terms of the English terms or acronyms that appear in parentheses. This approach is based on the linguistic characteristics of Vietnamese. We propose a set of efficient rules for extracting and filtering English-Vietnamese bilingual terminologies from monolingual Vietnamese corpora.

3 PROPOSED METHOD

In this section, we present our method for automatically extracting English-Vietnamese bilingual terms from Vietnamese documents.

In Vietnamese specialized documents, when writing a term, especially a novel term, people often put after this term its corresponding English term in parentheses "()". In this study, we focus on extracting bilingual pairs of terms that include an English term in parentheses and its corresponding Vietnamese terms. The main problem to solve is to determine which fragment of Vietnamese text is a term corresponding to the English term in parentheses.

The proposed method is to use an unsupervised method for determining Vietnamese terms and a set of rules to automatically extract English-Vietnamese bilingual terms. First, we collect candidates from the text and then apply screening rules to decide whether a candidate is really a term or not, and extract English-Vietnamese bilingual terms. The diagram of the process of this method is shown in Figure 1.

The selection of candidates, language features and rules will be detailed in the subsections below.

3.1 Candidates selection

The following remark about the writing style found in Vietnamese documents is also true for several languages, as shown in [4]. When writing a Vietnamese term, especially a new one, the corresponding English term is often written immediately afterwards, enclosed in parentheses. For example, this Vietnamese sentence "Bác sỹ chỉ định chụp cắt lớp vi tính (CT scan)" is equivalent to the English sentence "The doctor appointed a CT scan". In the Vietnamese version, the term "CT scan" in parentheses is used for reference of the Vietnamese term "chụp cắt lớp vi tính".

From the observation above, we proceed to extract bilingual term candidates who are pairs of:

- An English term or acronym found in parentheses;
- Its Vietnamese translation which is a phrase in front of the brackets.

Consequently, in the first step, for each text in parentheses, we use an English monolingual dictionary with more than 370.000 entries to verify if that text is English. If yes, the text is selected as an English term candidate.In addition, we also consider each string in capital in parentheses as a possible acronym, so it can be a candidate of an English acronym, for example "WHO", which stands for "World Health Organization".

For each English term candidate found, we determine the candidate of its translation in Vietnamese. As a simple but effective rule, we choose to consider the first punctuation sign (":" "," ":"...) before the left bracket of the English candidate to be the starting point for extracting the Vietnamese candidate text.

For example, given the sentence "*Thông thường những vi khuẩn có ích (lactobacilli) sẽ áp đảo số lượng vi khuẩn gây hại (anaerobes)*", we get two following candidate pairs in the format of <Vietnamese term candidate> (<English term candidate>):

(1) *Thông thường những vi khuẩn có ích* (lactobacilli)
Normally useful bacteria
(2) *sẽ áp đảo số lượng vi khuẩn gây hại* (anaerobes)
will overwhelm harmful bacteria

Another example is the sentence "*Bệnh nhân được chỉ định tiến hành một số xét nghiệm như: xét nghiệm công thức máu toàn bộ (CBC), xét nghiệm nước tiểu (UA).*", we can get two following candidate pairs:

(1) *xét nghiệm công thức máu toàn bộ* (CBC)
Complete blood count test
(2) *xét nghiệm nước tiểu* (UA)
urine test

We introduce now some language features that will be used to deal with the selected pairs of texts.

3.2 Language information

In the next steps, we make use of the following language features in order to remove bad candidates or to adjust the Vietnamese text fragments for obtaining Vietnamese terms:

- The mutual information (MI) [16] is used to determine whether a phrase in brackets is actually a term or not, and this information also helps to extract the corresponding Vietnamese term from the Vietnamese context in front of an English term.
- The expected number of Vietnamese syllables per English word in a English-Vietnamese translation. The Vietnamese term candidates which are too long or too short in syllables comparing to this ratio will be considered to be adjusted for a more reasonable length.
- Information about word categories such as conjunctions, prepositions of positions, stop words, number, *etc.* are used to cut off redundancies in Vietnamese contexts.
- A Vietnamese dictionary with more than $40,000$ words is used to check if the Vietnamese candidate extracted is actually Vietnamese text.

The language information used is all basic information that can be extracted directly from the raw corpora.

After the first step, we obtain a list of English terms or acronyms, and for each term we get a Vietnamese text which is the translation candidate for that term. In the next subsection, we establish some rules for processing these Vietnamese candidates in order to find English-Vietnamese bilingual term pairs.

3.3 Extracting English-Vietnamese bilingual terms

The extraction of English-Vietnamese bilingual terms is done through the application of the rules on candidates to extract the Vietnamese term corresponding to the English term in brackets. The rules are described in detail below.

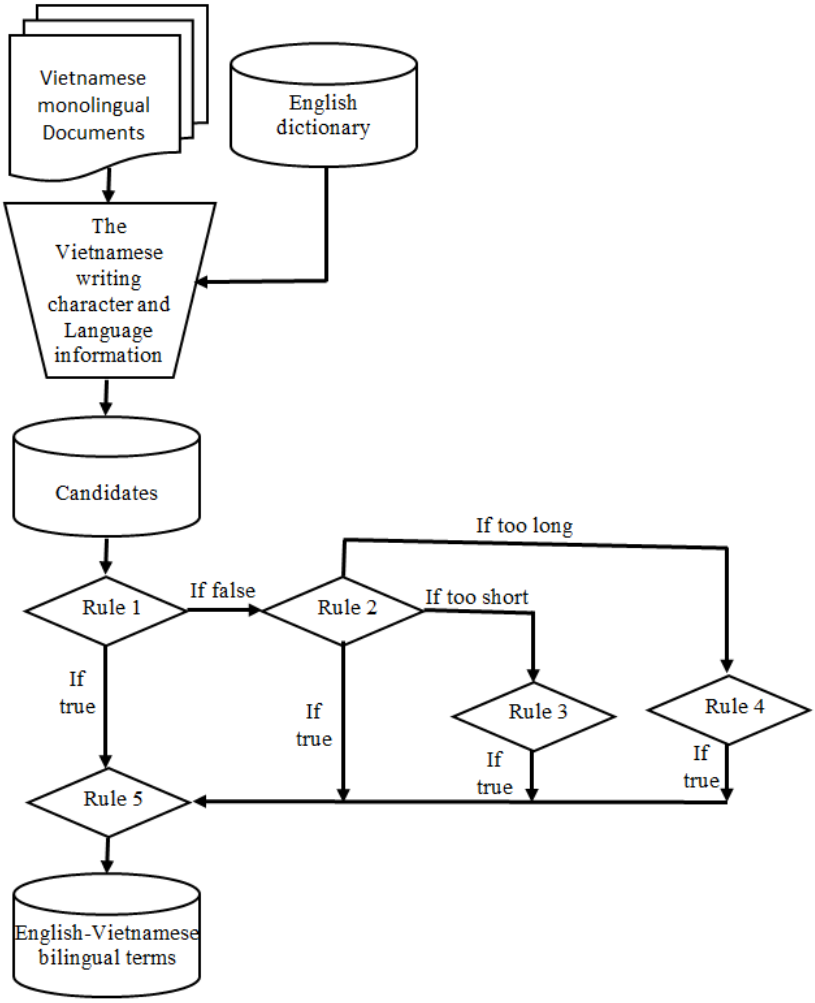


Figure 1: Schematic view of automatically extracting English-Vietnamese bilingual terms from Vietnamese documents.

- Rule 1: Two types of terms are extracted and considered as final result as they have a high degree of certainty for being the searched term:
  - Texts placed in double quotation marks " ", or single quotation mark ' ';
  - Terms which are organization names, based on the word capitalization.

  However, not many candidates satisfy this rule.
- Rule 2: This rule relies on mutual information (MI) to determine whether a candidate is a term or not and also determine if the Vietnamese term corresponding to the English term.
  - According to information theory, if a candidate is a term, the term will not be changed by context, so there will be a common string present inside the context in front of the English term.
  - If no common string found in the contexts of a same English term, or we find a common string with low frequency, the candidate is not a term.
  - The example below shows the 10 preceding contexts of the English candidate "*video*"
    - Burpee
    - Burpee
    - Hít đất
    - Hít đất
    - Hướng dẫn thực hiện động tác Superman
    - Hướng dẫn thực hiện động tác Body Saw
    - Hướng dẫn thực hiện động tác Superman
    - Hướng dẫn thực hiện động tác Body Saw
    - Hướng dẫn thực hiện động tác Lunge Kick
    - Bài tập thể dục Plank nghiêng vặn người

    Those different contexts don't allow to find a Vietnamese term corresponding to the English term "*video*".

  - The example below shows 10 preceding contexts of the candidate *ALS, i.e. "Amyotrophic Lateral Sclerosis"*
    * Điều trị **bệnh xơ cứng teo cơ cột bên**
    * Nguyên nhân gây **bệnh xơ cứng cột bên teo cơ**
    * Những ai thường mắc phải **bệnh xơ cứng cột bên teo cơ**
    * Những triệu chứng và dấu hiệu của **bệnh xơ cứng cột bên teo cơ**
    * Những phương pháp nào dùng để điều trị **xơ cứng cột bên teo cơ**
    * Những yếu tố nào làm tăng nguy cơ mắc **bệnh xơ cứng cột bên teo cơ**
    * Những kỹ thuật y tế nào dùng để chẩn đoán **bệnh xơ cứng cột bên teo cơ**
    * Những thói quen sinh hoạt nào giúp bạn hạn chế diễn tiến của **bệnh xơ cứng cột bên teo cơ**

    From these contexts, we output **bệnh xơ cứng cột bên teo cơ** as the Vietnamese term candidate for the English term *ALS*.
  - If a common string is found inside the contexts of the same English term, and its frequency is greater than a threshold α, then the candidate is a term and the common string between contexts is likely to be the corresponding Vietnamese term.
  - We set the threshold $\alpha = 0.75$, although this threshold will be adjusted if the extracted term is too long or too short compared to the ratio between the word lengths of the English text and its Vietnamese corresponding text candidate.
  - In this study, we aim to extract the most common terms, but this method can also be expanded to find synonymous Vietnamese terms (the same reference to an English term), for example "bỏng rạ" or "phỏng rạ" or "thủy đậu" (*Chickenpox*).

Rule 2 is the main rule in our method, it works effectively if the data is rich enough (high frequency and diverse in context). However, in addition to terms with many occurrences, there are also terms that have not many occurrences or have several translations. This leads to the extraction of the term incorrectly, too long or too short. The following rules are used to standardize the result of the term extracted from Rule 2.

- Rule 3: Applied to Vietnamese terms extracted with too short length compared to the length ratio between English words and Vietnamese words. As discussed above, this problem is caused by synonymous terms with different expressions, so the common string may be evenly distributed among instances.
  This rule performs a descending of the threshold α to find a term that is more appropriate. For example:
  - Consider 3 following candidate pairs: "Điều trị cho bệnh tế bào mast hệ thống (SM)"; "Bác sĩ sẽ chẩn đoán tăng dưỡng bào hệ thống (SM)"; "Việc nghi ngờ chẩn đoán bệnh tế bào mast hệ thống (SM)".
  - When $\alpha = 0.75$, the extracted term is *"hệ thống"*.The ratio of the word length between this Vietnamese term and its English term is 0.5. It's too small, because this ratio is popular at around 1.57, so we will gradually lower α threshold to find a longer Vietnamese term. When $\alpha = 66.66$, the ratio of the word length between this Vietnamese term and its English term is 2.0. As this ratio is good enough, we stop lowering α. And we get the Vietnamese term: *"bệnh tế bào mast hệ thống"*.
- Rule 4: Applied to extracted terms that are too long because the number of contexts is not big enough to clearly distinguish the boundaries of the term. Rule 4 is divided into 2 cases:
  - Rule 4.1: Increase the α threshold to reduce the length of the common string. If this does not help reduce the length of the term candidate, then apply Rule 4.2.
  - Rule 4.2: Use linguistic information about word categories and stop words to shorten common strings by identifying words such conjunctions, prepositions, numbers, *etc.* around the appropriate location corresponding to the ratio of English words and Vietnamese words. For example, the candidate "*tiếp xúc với khí ra-đông (radon)*" has the following part-of-speech annotation "*tiếp_xúc/V với/E khí/N ra-đông/N*". The word "*với*" having the part-of-speech "*E*" (preposition) is a sign to cut off the length of the context.
- Rule 5: It is actually a post-processing step, removing characters are not letters or remove stop words which are at the beginning of terms, or remove terms which are not Vietnamese.

The rules are applied sequentially. First, Rule 1 is applied on all the selected candidates to extract terms with a high degree of certainty. The remaining candidates will apply Rule 2, terms that are too long or too short compared with the length ratio between English words and Vietnamese words will apply Rule 3 or 4 to get better results. Finally, the results will apply Rule 5 to be standardized.

For example:

- Given two candidates: *"Có thể cần chụp Scan phóng xạ (Radionuclide scans)"; "Bác sỹ khuyên bạn cần chụp Scan phóng xạ (Radionuclide scans)"*
- For these candidates, the extracted Vietnamese term is *"cần chụp Scan phóng xạ"*, where the word *"cần"* is a stop word at the beginning of the candidate term, so it is removed from the candidate term. Finally We get the Vietnamese term *"chụp Scan phóng xạ"* corresponding to the English term "Radionuclide scans".

4 EXPERIMENTS

In studying the rules, we conducted experiments on two corpora. The first experiment deals with medical texts collected

from the Web. The size of this raw text corpus is about 334 MB. The second experiment is carried out on Vietnamese texts from Wikipedia. This second dataset was retrieved on Septtember 1st, 2018, and includes about 850 MB of raw text[1].

Table 1 shows the statistics of the English term candidates appearing in parentheses. For the medical corpus, the number of bilingual term candidates extracted is 910 with a total of 9779 occurrences. For the Vietnamese Wikipedia texts, we have extracted 6590 English candidates, with a total of 28861 occurrences. 152 English term candidates extracted from the medical corpus and 1724 English term candidates from Wikipedia are removed after the application of the rules presented in the previous section.

Table 1: Statistics of English term candidate extraction

Data	Medical corpus	Wikipedia
# Term candidates	910	6590
# Occurrences	9779	28861
# Removed candidates	152	1724

The number of rule applications to the medical bilingual term candidates is shown in Table 2.

Table 2: Number of rule applications on medical data

Rule 1	Rule 2	Rule 3	Rule 4
11	899	201	243

The first rule is applicable on 11 term pairs. The 899 remaining term pairs are then processed with the application of Rule 2. Among these 899 term pairs, Rule 3 is applied to 201 pairs and Rule 4 is applicable to 243 other pairs. Note that Rule 3 and Rule 4 are used to smooth the results obtained once Rule 2 applied. All final term pairs must be processed by the application of Rule 5 as a standardization step.

Similarly, the number of rule applications on Wikipedia data is presented in Table 3.

Table 3: Number of rule applications on Wikipedia data

Rule 1	Rule 2	Rule 3	Rule 4
855	5735	1112	1898

We conducted an assessment of the resulted bilingual term candidates by manually checking 500 randomly selected term pairs. The results are presented in Table 4.

From the statistics in Table 1, we can note that the average number of instances of a term in the medical corpus is much higher than in the Wikipedia data (about respectively 10 and 4 occurrences for each medical term and Wikipedia term). This is quite natural because Wikipedia text is written in a different style in comparison with other documents. A corpus like the medical corpus focuses on a specific topic, so the instance density of a term is higher. We also note that the explanation involving the appearance of English terms in the Vietnamese Wikipedia corpus has frequently a more complicated language context, which results a higher error rate in term pair detection.

[1]https://dumps.wikimedia.org/backup-index.html

Table 4: Evaluation

Data	# Evaluated terms	# Correct terms	Accuracy
Medical corpus	758	626	82.6 %
Wikipedia	500	371	74.2 %

Table 5 shows a analysis of the validated bilingual terms. Among the 626 correct term pairs from the medical corpus, there are 355 acronyms and 40 terms found in a medical and general English-Vietnamese dictionary containing 90,000 English entries. For the 371 correct bilingual terms extracted from Wikipedia, there are 105 acronyms and 11 term pairs which are present in the above dictionary. These numbers shows that a large proportion of the English-Vietnamese bilingual terms extracted and validated from our monolingual corpora is very useful.

Table 5: Resulted term analysis

Data	Medical corpus	Wikipedia
# Correct terms	626	371
# Acronyms	355	105
# Existing in dictionary	40	11

5 CONCLUSION

We have presented a new rule-based method for extracting English-Vietnamese terminologies from mostly monolingual Vietnamese corpora. Experiments conducted on a medical corpus and on Vietnamese texts extracted from Wikipedia show promising results, which are respectively 82.6% and 74.2% of accuracy. These bilingual term candidates can be manually revised to obtain gold bilingual terms which are very useful in terminology study on one hand and in NLP applications like machine translation or cross-language information retrieval on the other hand.

This approach allows us to acquire English-Vietnamese bilingual term candidates in various specific domains by exploiting monolingual corpora from these domains. In the future, we will combine this approach with machine learning approaches in order to better filter the candidate terms.

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